

Tribes.AI: Building a next-gen automated timesheet solution to prevent daily losses of up to US\$9.1B

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About Tribes.AI

Leveraging Google
build its next-gen
activity-based time
platform, Tribes.AI
scale for future growth

Tribes.AI provides an automated activity-based time tracking solution that delivers accurate and timely reports to reduce the risk of human error, find unbilled hours, and link revenue earned to time spent.

Industries: Technology

Location: Australia

(<https://cloud.google.com/storage/>), [Compute](#)

Google Cloud results

- Speeds up app development through automation and microservices with GKE
- Builds a complex events identification model that can account for human behavior
- Enables new engineers to build quickly with solutions that have a shallow learning curve

Build
active
time
platform
MVP
autor
prod
month

Google Cloud (<https://cloud.google.com/>),
Kubernetes Engine

App Engine (<https://cloud.google.com/appengine/>)

Looker Studio

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Looker Studio

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Looker Studio
(<https://marketingplatform.google.com/about/data-studio/>)

"I feel it is a lot easier to build a product on Google Cloud. It's easier to have someone that doesn't understand Google Cloud say, look, go here, do this, spin up an instance or connect this with that. All of that together accelerates getting a product going."

—*Christian Bartens, CEO
and Co-founder, Tribes.AI*

There are a few things employees prefer not to do on a day-to-day basis, like manually filling in

timesheets. Understanding this behavior is an observation of human nature itself, and when people dislike something, they put it off to a later time, do a bad job, or forgo doing it altogether. Then, when employees finally attempt to fill their timesheet on a Friday while trying to account for every hour of the past week, their timesheets are bound to be filled with errors.

Timesheets play a vital role for businesses like consultancies as they are key to billing customers accurately. With this in mind, Christian Bartens, CEO and Co-founder of Tribes.AI (<https://www.tribes.ai/>), is on a mission to turn timesheets into a revenue generator not just for consultancies, but any business. Founded in 2017, Tribes.AI leveraged the Google for Startups Cloud Program (<https://cloud.google.com/startup>) to build a next-generation, activity-based time tracking (ABT) platform that eliminates human error, delivers accurate timesheets, and finds previously unbilled hours.

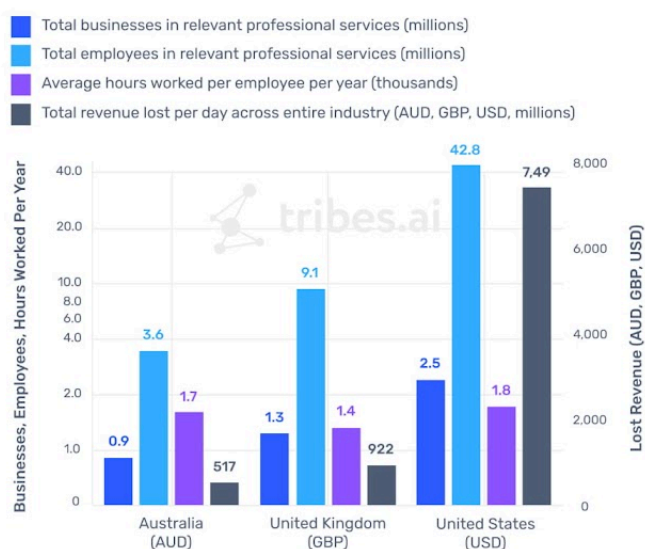
"We started estimating time loss, and based on the research that we've done, we found that 20% to 30% of billable time is actually not recorded. Most service businesses operate on a profitability margin of about 10%. So imagine if 20% to 30% of your work is not recorded and invoiced, then you're really harming your business," explains Bartens.

Tribes.AI research reveals

(<https://www.tribes.ai/global-timesheets-benchmark/report>)

that this problem costs businesses in the US, UK

and Australia up to US\$9.1 billion per day or US\$51,000 per employee annually. Adopting a two-pronged approach, Tribes.AI is helping employees do away with tedious manual work and enabling employers to plug revenue attrition from flawed timesheets.



Inaccurate timesheets cost USD 9.1b per day

Building smart models with a distributed team amid a talent crunch

Barten's previous business venture was a Google Analytics (<https://analytics.google.com/>) premium partner in APAC as well as a Google Cloud partner (<https://cloud.google.com/partners>), so turning to Google Cloud to build and run his new ABT was a natural progression of that long standing relationship.

"I feel it is a lot easier to build a product on Google Cloud," he explains. "You can have someone that doesn't understand Google Cloud say, look, go here, do this, spin up an instance or connect this with that. All of that together accelerates getting a product going."

Tribes.AI realized that Google Cloud could help them address a tech startup's main challenges, which are cost and scalability. Moreover, they could achieve this amid a tech talent crunch because Google Cloud enables developers to start working and easily figure things out for themselves despite not having much experience with it.

The company's next-generation ABT uses a series of API integrations that pulls data from Google Workspace and other productivity platforms to look for events that an employee caused. An event could be opening and sending an email, posting a message, opening and editing a file, and much more, but the ABT cannot read the content of the event, protecting the privacy of employees.

The focus is the time spent for every event, multiplied by thousands of employees. The ABT uses these signals to determine how much someone worked, much like how a regression model operates, and aggregates the time spent on events at the employee, team, and client levels.

However, that data is also influenced by human behavior. Tribes.AI's models can account for outlier events like sending emails at 4 AM and distinguish between automated behavior, like [Google Sheets](#)

(<https://www.google.com/sheets/about/>) automatically saving a file every three minutes, versus human intervention, such as manually saving a file.

"With
Kubernetes
Engine you can
build something
that can
automatically
scale when you
become
successful. You
don't want to
just build
something,
launch it, then
realize that when
you sign 10 or
15 new clients,
you're not
processing
1,000 employees
anymore, but
you are
processing

10,000. You
want to build
everything right
away on a
scalable
platform."

—Christian Bartens, CEO
and Co-founder, Tribes.AI

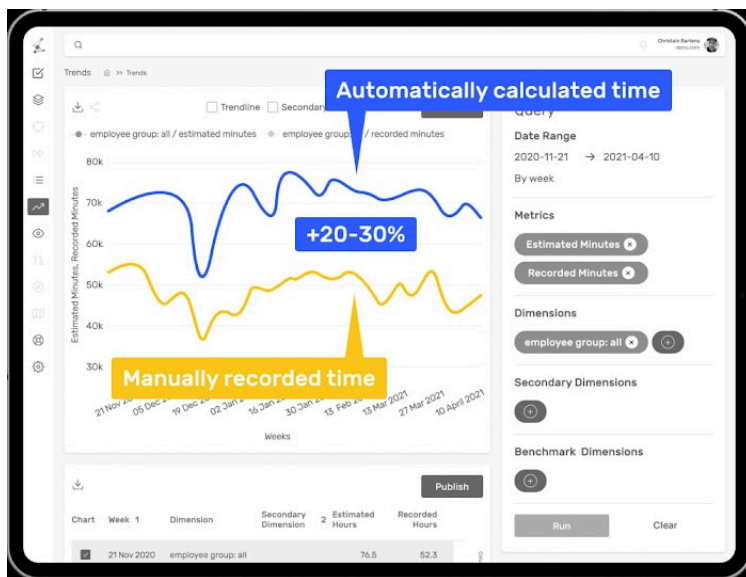
Getting the right foundation for scalability

On the back end, Google solutions like [Cloud Run](https://cloud.google.com/run) (<https://cloud.google.com/run>), [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) and [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>) work together to help Tribes.AI's team pull data on a daily basis and run its model once a week to process events, without any downtime. [Cloud Key Management](https://cloud.google.com/security-key-management) (<https://cloud.google.com/security-key-management>) is also used to store and encrypt clients' proprietary APIs.

With no capital for office space and physical servers, Google Cloud also proved vital for the company's teams to work collaboratively across multiple geographies. Tribes.AI has a distributed team with marketers in Nigeria, developers in India, and data scientists in South America.

Its developers can also rely on Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) to manage and deploy microservices and automate when they are switched on and off, thereby keeping costs in check for the startup.

However, Bartens reveals that another big reason why Kubernetes Engine is important for Tribes.AI is because of the company's plans to scale. "You want to build something that can automatically scale when you become successful. You don't want to just build something, launch it, then realize that when you sign 10 or 15 new clients, you're not processing 1,000 employees anymore, but you are processing 10,000. You want to build everything right away on a scalable platform," he says.



*Automatically Calculated Time vs Manually Recorded Time
App*

Helping drive the future of hybrid work

Tribes.AI (<https://www.tribes.ai/>) is uniquely positioned to support businesses, especially after the spike in remote work due to the pandemic. As more and more companies adopt permanent hybrid working arrangements for their employees, the Tribes.AI platform can help support managers by providing transparency for team productivity levels, and insights on where time is being spent to help enable work life balance. And for employees, the platform can help them adopt better time management practices at home, creating healthier and more productive behaviors in prioritizing their time.

Tribes.AI has signed five trial customers since its ABT launched in April 2022, and the company is ready to scale its product. Bartens is looking to further optimize costs during this process, saying that moving from App Engine (<https://cloud.google.com/appengine>) to Cloud Run has been more cost-effective when transitioning to a larger scale.

"There's a
Google Cloud
tool for every
step of our
maturity phase.
We might start
with App Engine

because it's
super easy,
everybody
knows how to
use it, and
quickly get
something done.
But then there's
something else
that we can
progress to,
should we
decide to focus
on scale and be
more efficient in
how we allocate
money and
resources."

—Christian Bartens, CEO
and Co-founder, Tribes.AI